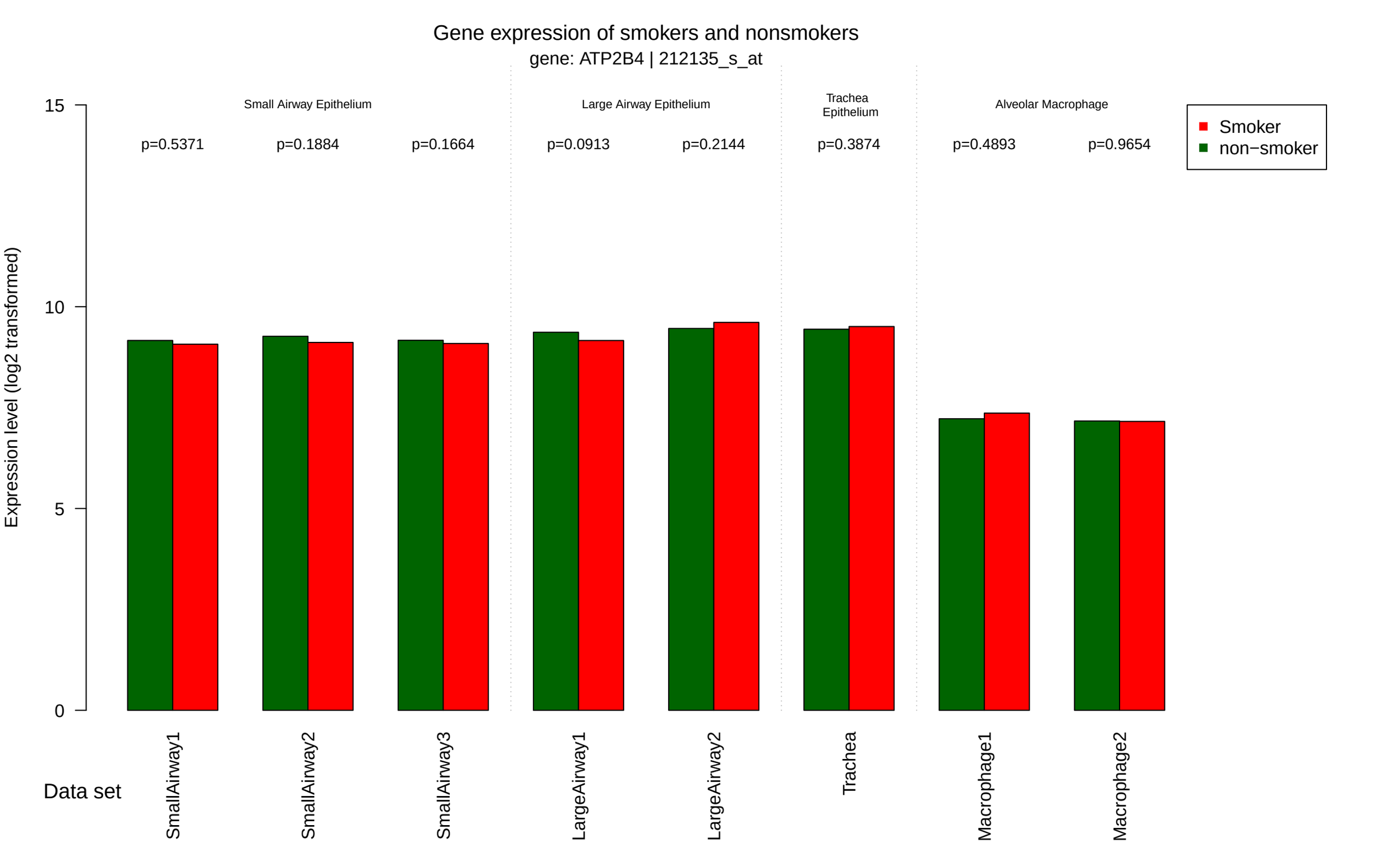
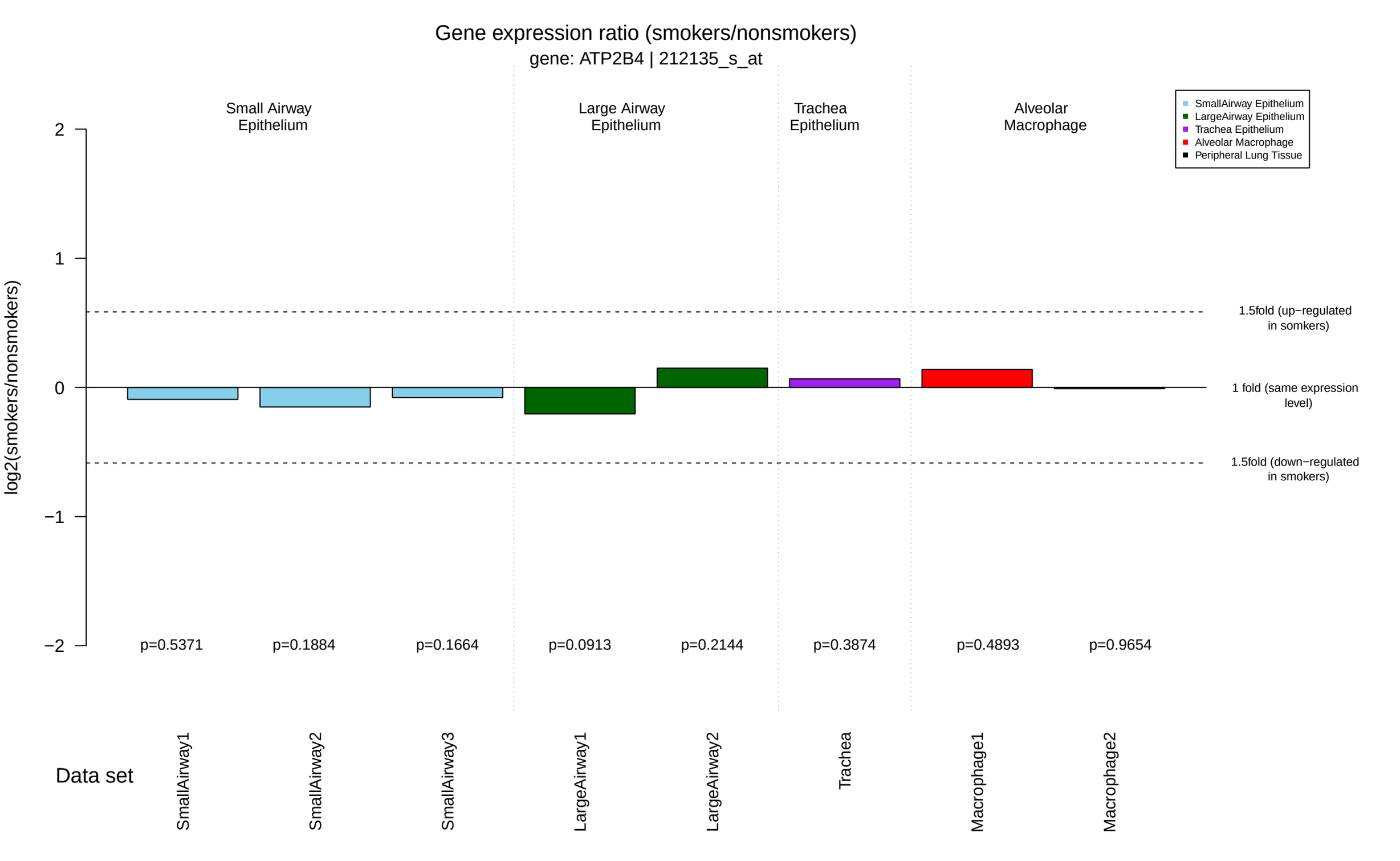
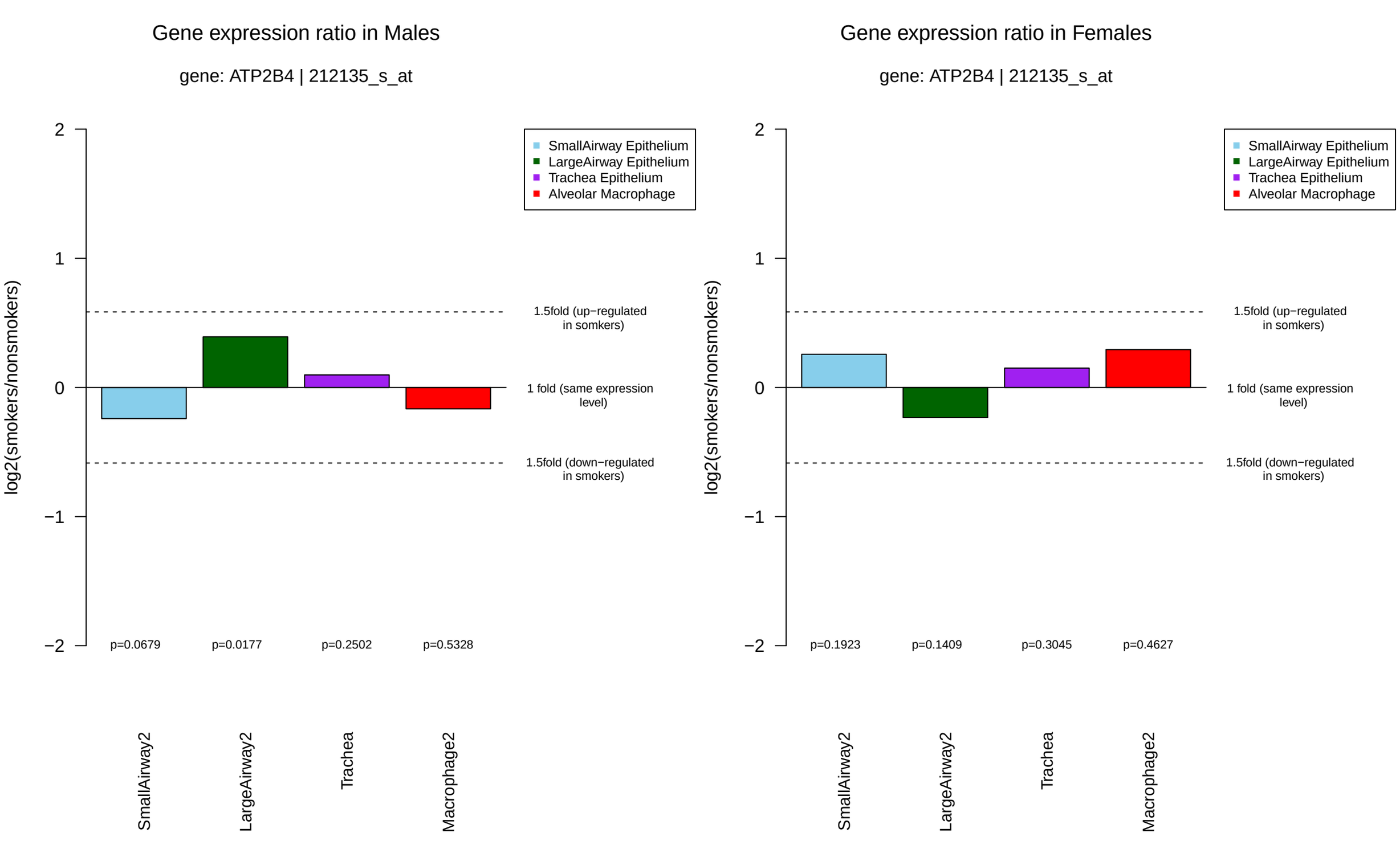
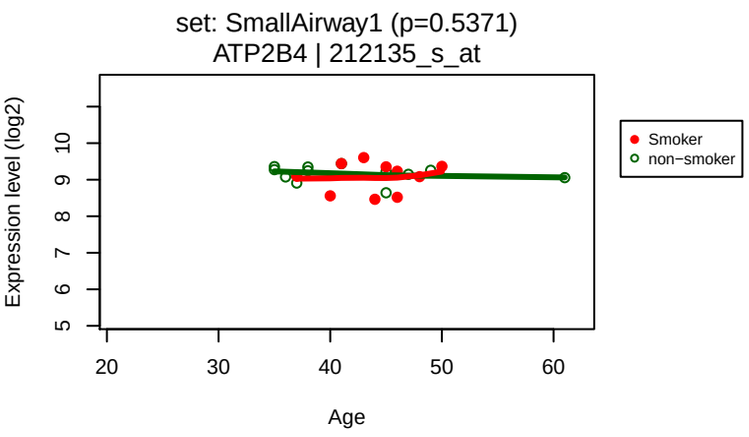
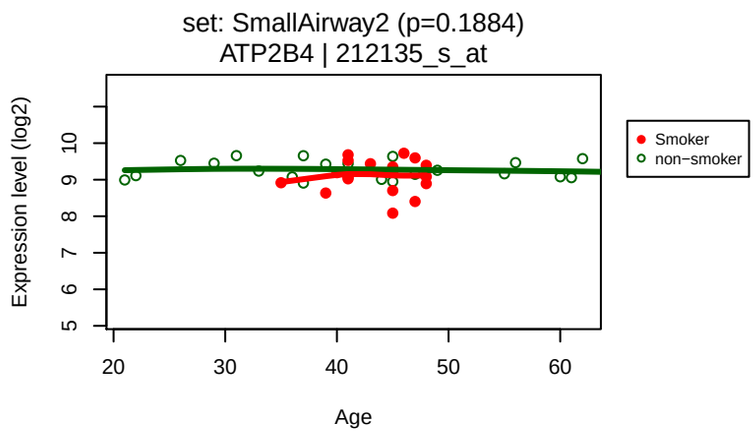
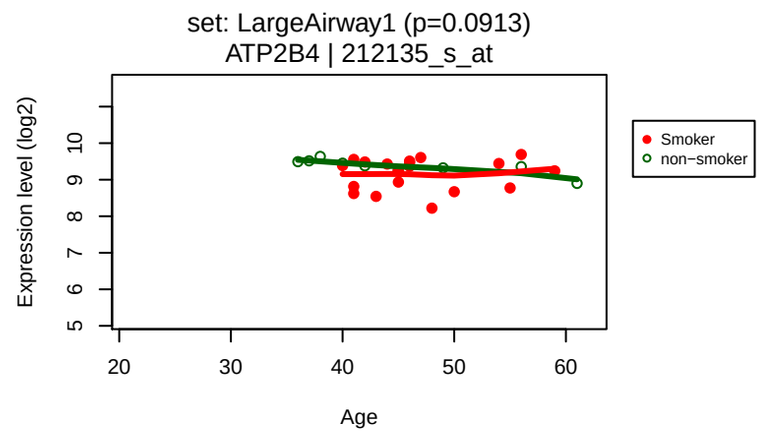

Gene expression of smokers and nonsmokers
gene: ATP2B4 | 212135_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5371
p=0.1884
p=0.1664
p=0.0913
p=0.2144
p=0.3874
p=0.4893
p=0.9654
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

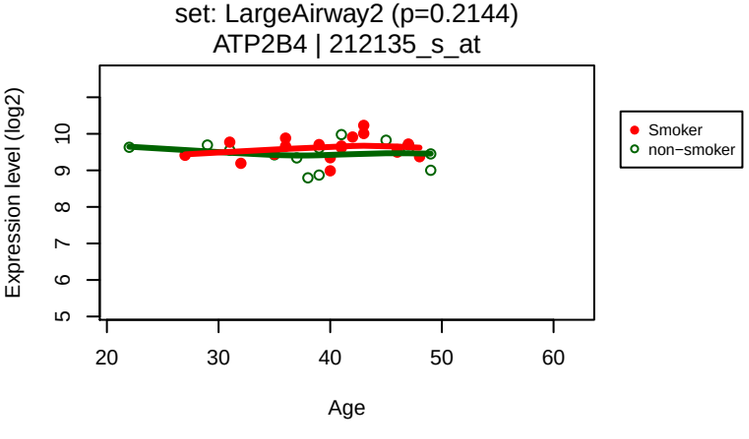

Gene expression ratio (smokers/nonsmokers)
gene: ATP2B4 | 212135_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5371
p=0.1884
p=0.1664
p=0.0913
p=0.2144
p=0.3874
p=0.4893
p=0.9654
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

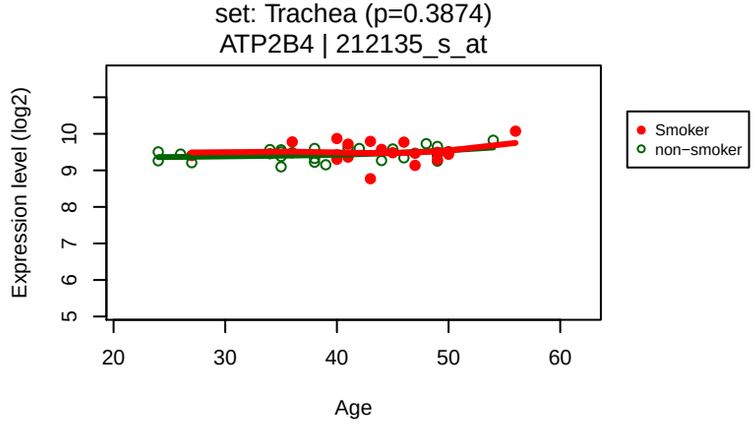

Gene expression ratio in Males
gene: ATP2B4 | 212135_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0679
p=0.0177
p=0.2502
p=0.5328
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2B4 | 212135_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1923
p=0.1409
p=0.3045
p=0.4627
SmallAirway2
LargeAirway2
Trachea
Macrophage2

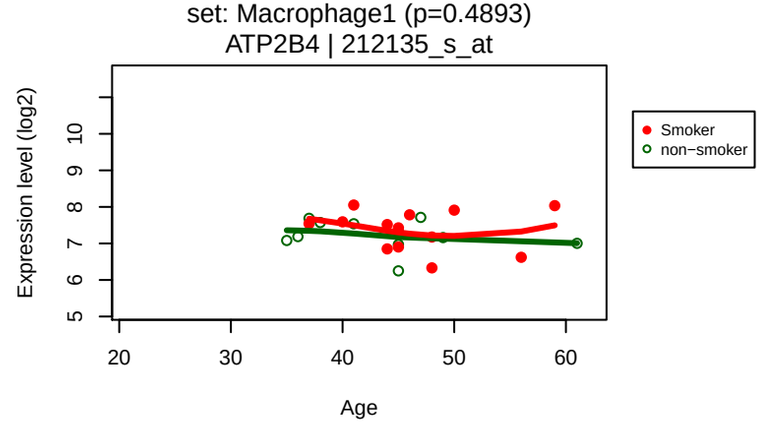

set: SmallAirway1 (p=0.5371)
ATP2B4 | 212135_s_at
Expression level (log2)
Age
Smoker
non−smoker

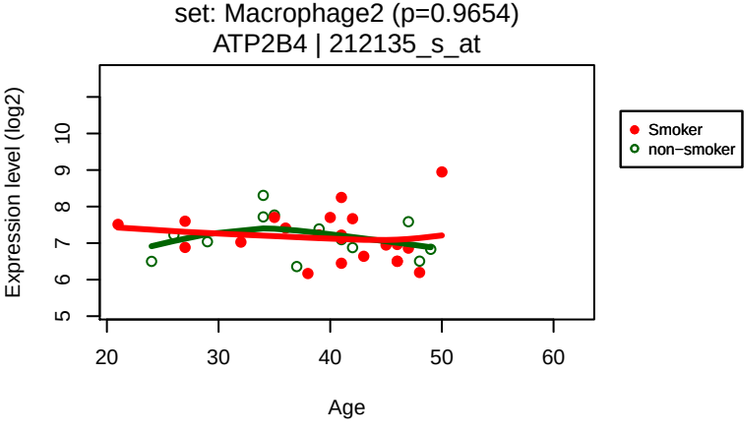

set: SmallAirway2 (p=0.1884)
ATP2B4 | 212135_s_at
Expression level (log2)
Age
Smoker
non−smoker

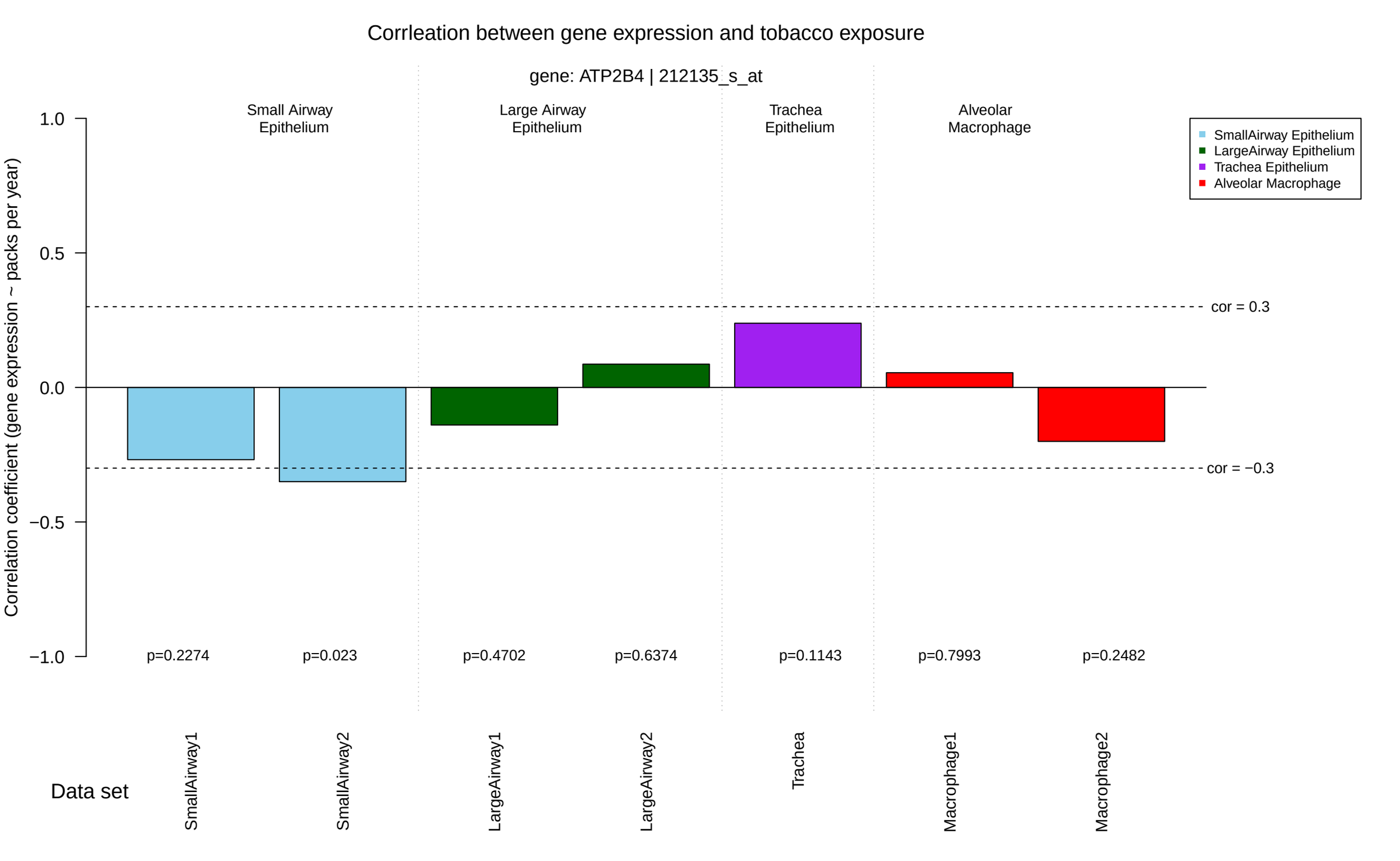

set: LargeAirway1 (p=0.0913)
ATP2B4 | 212135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2144)
ATP2B4 | 212135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3874)
ATP2B4 | 212135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4893)
ATP2B4 | 212135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9654)
ATP2B4 | 212135_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2B4 | 212135_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2274
p=0.023
p=0.4702
p=0.6374
p=0.1143
p=0.7993
p=0.2482
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2